Varala Akshay Kumar

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.18
NARAYANA JR COLLEGE	Class XII (TSBIE)	2023	97.00%
DR KKR GOWTHAM CONCEPT SCHOOL	Class X (SSC)	2021	10.00%

PROJECTS

Machine Learning for Credit Default Prediction on LendingClub's P2P Platform | Business Analytics

- Built an end-to-end credit risk pipeline on **2.26M LendingClub loan records**, engineering 99 features from 151 raw variables and applying **VIF filtering and Lasso regularization** to reduce to 77 predictive features.
- Trained and benchmarked **six classifiers** (Logistic Regression, Random Forest, **XGBoost**, SVM, and others), correctly applying **SMOTE inside cross-validation folds** to eliminate data leakage, achieving **0.7330 AUC-ROC** -- a **7.8% improvement** over LendingClub's own grading system.
- Used **threshold optimization** to raise defaulter recall from 10.8% to 62.0%, deploying the model on 912,569 active loans to flag 32.1% as high-risk with an estimated **$220M in prevented losses**.

Agentic-RAG-MCP | Retrieval-Augmented Generation / Agent Tooling

- Built a **local-first RAG retrieval engine** exposed as a **Model Context Protocol (MCP)** server, enabling any MCP-compatible agent (opencode, Claude) to ground responses in private documents without loading a redundant LLM.
- Implemented **instruction-aware embeddings** using **Qwen3-Embedding (GGUF via llama.cpp)** with separate document/query prefixes, backed by a local **Qdrant vector store** with metadata filtering and a pluggable embedding registry (SentenceTransformers, OpenAI, Cohere).
- Designed a modular, ABC-based pipeline (converters, chunking, retrieval, reranking) with sandboxed ingestion against path traversal and injection attacks, validated by **51/51 passing unit tests**.

Financial Decision Making for Stock Signals | Natural Language Processing :

- Developed a **3-class financial decision-making model** to predict BUY, HOLD, or SELL signals for Bitcoin (BTC) and Tesla (TSLA) using daily news, price, momentum, and financial data.
- Extracted sentiment features using **FinBERT** and engineered market indicators -- returns, volatility, momentum, and z-scored features -- using **7-day sliding windows** with chronological data splitting to prevent leakage.
- Built a **GRU-based sequence model** with asset embeddings and attention using asymmetric loss, achieving **50% test accuracy** and **0.4877 Macro-F1** on the financial signal classification task.

Defense Against Adversarial Attacks on Image Recognition | Computer Vision :

- Developed an **adversarially robust image classification pipeline** using **ResNet-18** on CIFAR-10 with **PGD-based adversarial training** and **Exponential Particle Swarm Optimization (ExPSO)** for structured pruning.
- Implemented **Gumbel-Softmax stochastic multi-expert routing** with compressed models at 30%, 50%, and 70% sparsity, achieving **47.75% PGD-20 robust accuracy**.
- Evaluated the defense using **FGSM, PGD-20, PGD-100, and AutoAttack**, achieving **45.20% AutoAttack robust accuracy** and up to **1.24× inference speedup**.

SKILLS

Programming Languages: Python, C++, C, SQL, RISC-V Assembly

Machine Learning & Deep Learning: PyTorch, XGBoost, scikit-learn, NumPy, Pandas, Matplotlib, OpenCV, FinBERT/Transformers, HuggingFace

Agentic AI & RAG: RAG, Agentic AI, MCP, LangChain, LangGraph, llama.cpp, Qwen3-Embedding

Databases & Vector Stores: MySQL, Qdrant

Tools: Git, GitHub, Jupyter Notebook, Google Colab, LaTeX, SolidEdge

RELEVANT COURSES

- **Artificial Intelligence:** Probability and Random Variables, Programming for AI, Matrix Theory, Foundations of Machine Learning, Deep Learning, Computer Vision, Natural Language Processing
- **Mathematics :** Calculus, Applied Statistics, Bayesian Statistics.
- **Computer Science:** Introduction to Programming, Data Structures, Algorithms, DBMS

POSITIONS OF RESPONSIBILITY

Hospitality & Logistics Head - Prayas RDC Club :

- Led a team of coordinators to manage hospitality and logistics for a rural outreach club focused on educating underprivileged children, overseeing coordinators for on-site teaching visits and campus learning activities.

Campus Tour Coordinator - Alumni Cell :

- Coordinated campus tours for visiting alumni, strengthening engagement between alumni network and students.

EXTRACURRICULAR

National Service Scheme (NSS) | Social Outreach

- Completed **120+ hours** of community service through **cleanliness drives and school teaching, tree plantation and community outreach initiatives** promoting education and environmental awareness.